

RAFFLES INSTITUTION

Higher 2

CANDIDATE
NAME

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GROUP

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BIOLOGY

Paper 3

9648/03

20 September 2011

2 hours

Additional materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, **hand in your essay SEPARATELY.**

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/13
2	/14
3	/13
4	/12
5	/20
Total	/72

This document consists of 11 printed pages.



Raffles Institution
Internal Examination

Answer **all** questions.

- 1 Fig. 1.1 below illustrates how a human protein, YS56, can be produced using bacteria.

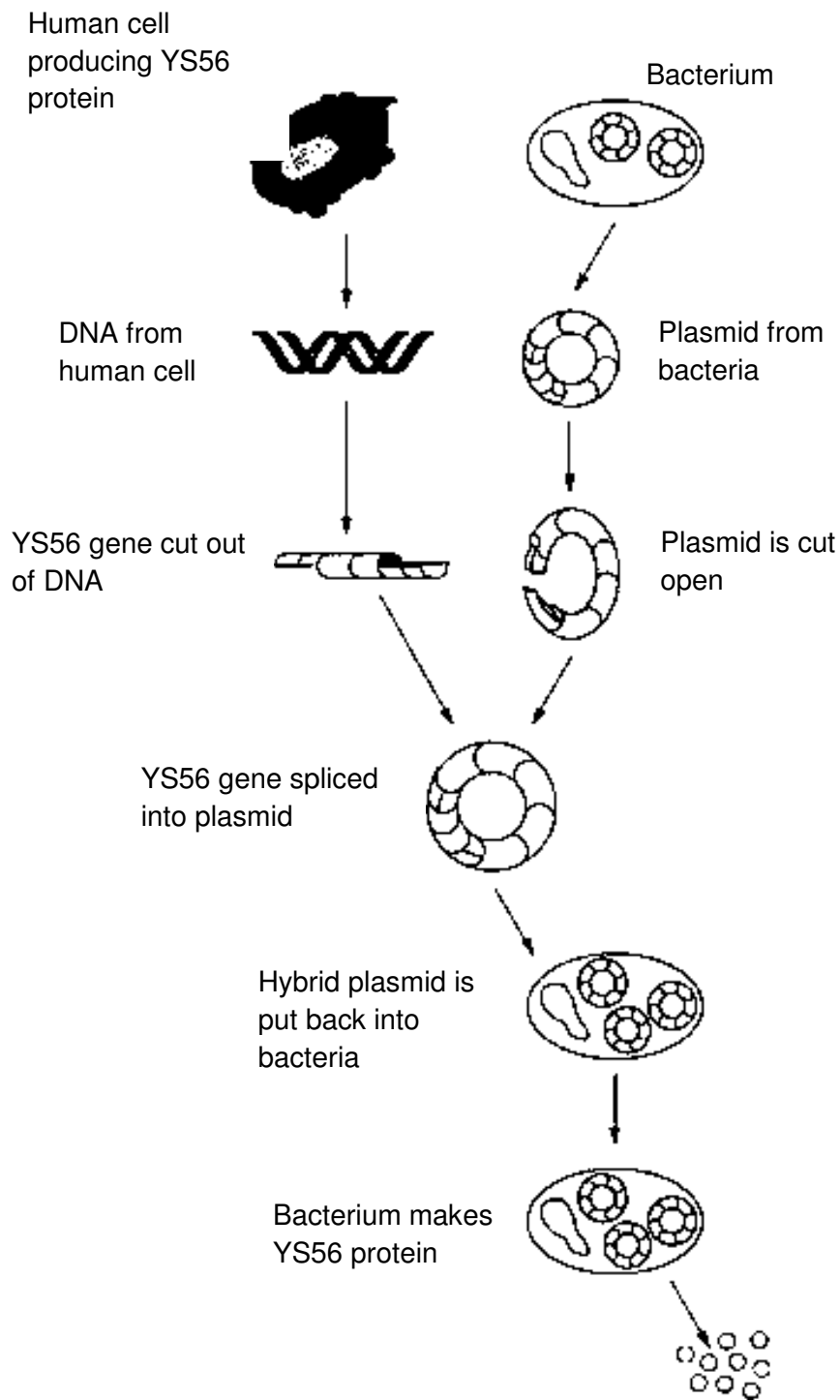


Fig 1.1

- (a) Restriction enzymes are used to cut the YS56 gene out from human DNA. Explain the mechanism of action of restriction enzymes. [3]

- (b) State the enzyme that is used to splice the YS56 gene into the plasmid, and describe the reaction it catalyses. [2]

- (c) Explain how the recombinant plasmid can be “put back” into the bacteria. [2]

- (d) Explain two considerations that must be fulfilled to ensure proper expression of the gene in the bacterium. [4]

- (e) Suggest the advantage of using yeast cells instead of bacteria as a host for the production of human proteins. [2]

[Total: 13]

2 Restriction fragment length polymorphism, RFLP, refers to differences in fragment length generated by restriction enzymes. These differences are used to detect variations in genomes. The technique is widely used in molecular biology.

- (a) Fig 2.1 below shows an autosomal dominant disease segregating in a family. Genomic DNA from each family member was assayed for RFLPs for *Eco*RI sites, on Southern blots using probes derived from the gene.

The results for all individuals, except those for the foetus, are shown below the pedigree chart. The sizes of the DNA fragments obtained are as indicated.

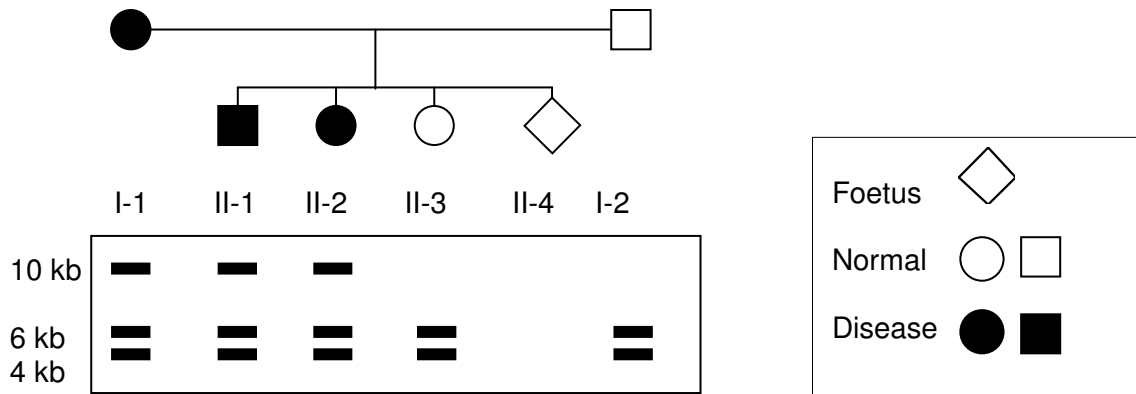


Fig 2.1

- (i) Sketch a suitable diagram to illustrate the region of genomic DNA recognised by the probe in individual I-1. Show the positions of *Eco*RI sites and the distance between them in kb. [1]

- (ii) Which RFLP fragment did the mother (I-1) contribute to the diseased children? [1]

- (iii) What is the probability of the foetus having the disease? Explain your answer. [2]

- (b) A different study was carried out on another family. Fig. 2.2 below shows a Southern blot with an RFLP detected by using a probe for a sequence located on a sex chromosome. The variant exhibits two forms, one at 8 kb and one at 6 kb. The mother, father and daughter are normal.

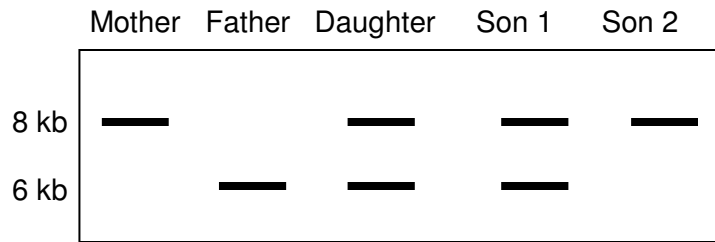
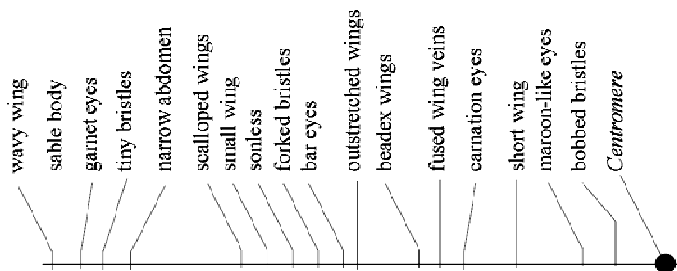


Fig 2.2

- (i) Which son is abnormal? [1]

- (ii) Explain how you arrived at your answer. [3]

- (c) Before the genome of an organism is sequenced, preliminary mapping is usually first carried out. Shown below is a chromosome map for the X chromosome of the fruit fly *Drosophila melanogaster*.



- (i) What is a chromosome map? [1]

- (ii) Explain how restriction fragment length polymorphism, RFLP, can be used with linkage mapping to construct such a chromosome map. [4]

- (iii) Suggest one limitation scientists would have faced if they used linkage mapping techniques to map the human genome. [1]

[Total: 14]

- 3** Round-Up Ready™ soybeans are genetically modified to be resistant towards Round-Up™ herbicide, a glyphosate-based herbicide that normally inhibits an important enzyme in the aromatic amino acid synthesis pathway in plants. This allows farmers to freely apply Round-Up™ to destroy weeds without affecting the Round-Up Ready™ soybeans. Round-Up Ready™ soybeans may be produced by first transfecting soybean callus cells with the resistance gene from *Agrobacterium*.

- (a)** To obtain a callus culture, suitable explants must first be chosen. Suggest a suitable explant and explain your choice. [2]

- (b)** Outline the steps in obtaining a callus culture from an explant. [3]

- (c)** State two advantages of using callus cells in producing genetically modified soybeans. [2]

- (d)** Explain how cells are screened to check which ones have successfully taken up the resistance gene. [2]

- (e) The desired callus cells can then be induced to form plantlets. State the ratios of plants growth regulators required for this. [2]

- (f) Despite offering benefits to farmers and consumers in terms of better yield, various concerns have also been raised regarding the use of Round-Up Ready™ soybeans. State one concern and suggest a way of overcoming it. [2]

[Total: 13]

Planning question**4**

Plants carry out photosynthesis under sunlight or white light that is made up of different wavelengths. Different wavelengths may affect the rate of photosynthesis to different extents. An action spectrum shows the effect of the wavelength of light on the rate of photosynthesis.

Design an experiment, using a suitable aquatic plant, e.g. *Elodea* sp. to investigate:

The effect of different wavelengths of light on the rate of photosynthesis of an aquatic plant.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you may choose to use:

- Different coloured filters of known wavelengths and maximum transmission
- Large beaker
- Filter funnel
- Delivery tube
- Gas syringe
- Stopwatch
- Thermometer
- Lamp
- Sodium bicarbonate
- Light meter

[Total: 12]

Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labeled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b) etc, as indicated in the question.

- 5 (a)** Describe the viral-mediated approach to gene therapy to treat SCIDs and explain why the *ex vivo* approach was adopted. [8]
- (b)** Discuss the factors which prevent gene therapy from becoming an effective treatment. [6]
- (c)** Describe the unique features of stem cells. Explain the differences between stem cells and cancer cells. [6]

[Total: 20]